

The SC-SCOPE endpoint floor is not tight – sharpening it via the

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The SC-SCOPE all-orders endpoint ($I = 2 \times 10^{-3}$) did not close by third-order means: the joint incompatible-pairing recovered only $\times 0.905$ (scscope-joint-pairing v1.0), which named the SOLE remaining route as “a larger second-order STEP-5B endpoint floor: if the de-thinned floor were $\gtrsim \times 3.9$ the paired joint would close.” This note completes that route by observing that the floor $\rho = 2.58$ rests on a LOOSE additive-energy bound, and the additive-energy result of the DR-2 programme (R-025/R-026) supplies the tighter constant.

2. The floor composition

The de-thinned STEP-5B closure floor (robustness_mu2_step5b_remargin) is

$$\rho = \frac{K_{\text{budget}}}{K(n_{\text{pack}})}, \quad K_{\text{budget}} = \frac{4(1 - a_0) m}{(\lambda' I)^2 J_{\text{eff}}}, \quad K(n_{\text{pack}}) = 8 + 4\sqrt{14} \sqrt{n_{\text{pack}}},$$

where $K(n_{\text{pack}})$ is the kappa-balanced additive-energy bound and $n_{\text{pack}} = 16/\theta_{\text{min}}^2$. At the endpoint (anchor $\mu^2 = 0.005$): $n_{\text{pack}} = 40.7$, $K(n_{\text{pack}}) = 103.5$, $K_{\text{budget}} = 266.7$, so $\rho = 2.58$ (reproducing the AddE $\times 2.6$).

3. The bound is loose at the small endpoint

$K(n) = 8 + c_R \sqrt{n}$ with $c_R = 4\sqrt{14} \approx 15$ is the UNCONDITIONAL triple-count bound ($E_+ \leq n^{5/2} \Rightarrow K = E_+/n^2 \leq \sqrt{n}$ up to the prefactor). At the endpoint $n_{\text{pack}} = 40.7$ this gives 103.5, which EXCEEDS the trivial Lemma-A additive-energy bound

$$\frac{E_+(Q)}{N^2} \leq 1 + T'(Q), \quad T'(Q) \leq |Q| \leq n_{\text{pack}} = 40.7 \quad \Rightarrow \quad 1 + T' \leq 41.7 < 103.5.$$

The large prefactor $c_R \approx 15$ makes the $\sqrt{n}$ bound overshoot the trivial $O(n)$ bound at this moderate n . The additive-energy constant K and $1 + T'$ are the SAME λ' -free quantity ($\langle F^4 \rangle - 4I^2)/I^2$ (via $\sum w^2 = \lambda'^2(\langle F^4 \rangle - 4I^2)$ and $\sum w^2 \leq K(\lambda' I)^2$), so $1 + T'$ is a legitimate, tighter substitute.

4. The sharpened floor closes the endpoint

Substituting the tighter constant, $\rho_{\text{lat}} = K_{\text{budget}}/(1 + T')$:

T'	$1 + T'$	ρ_{lat}	paired = $\rho_{\text{lat}}/2.872$
2	3	88.9	30.9
8	9	29.6	10.3
$n_{\text{pack}} = 40.7$	41.7	6.40	2.23
67	68	3.92	1.37
92	93	2.87	1.00

In the H-ADM-COH-separated regime $T' \leq n_{\text{pack}} = 41$, so $\rho_{\text{lat}} \geq 6.4 \geq 3.9$ and paired $\geq 2.23 > 1$ – the all-orders endpoint CLOSES. Break-even is $T' \leq 92$ (paired ≥ 1) and $T' \leq 67$ ($\rho \geq 3.9$). For the post-discharge lattice class the measured $T' \sim \text{tens}$ (≤ 48 for $N \leq 2040$, R-026), giving paired ≥ 1.90 – closes with margin. The third-order inflation $1 + \max[R_s + R_q] = 2.872$ is a separate third-cumulant quantity and is unchanged.

5. Reconciliation (PROVED) and the lift

The constant map is now pinned EXACTLY. In the STEP-5B convention $w_t = \lambda' \sum_{u+v=t} A_u A_v$ and, by the Parseval reformulation, $\sum_{t \neq 0} w_t^2 = \lambda'^2 (\langle F^4 \rangle - 4I^2)$, so the floor’s additive-energy constant is exactly $K = \sum_{t \neq 0} w_t^2 / (\lambda' I)^2$. The weighted Lemma A (R-027) gives, for its $t \neq 0$ part, $\sum_{t \neq 0} |w_t|^2 \leq T'(M) I^2$, hence

$$K = \frac{\sum_{t \neq 0} w_t^2}{(\lambda' I)^2} \leq T'(M)$$

where $T'(M)$ is the sum-circle richness of the mode set – NOT the looser kappa-balanced $8 + c_R \sqrt{n}$. Verified (scscope_constant_map.py, 3/3) on real $\pm k$ -symmetric sphere-lattice patterns: $K_{\text{floor}}/T' \leq 0.52$ and $w_0 = I$ exactly (so the $-4I^2$ subtraction is even more conservative, lowering K further). With $K \leq T' \leq n_{\text{pack}} = 40.7$ (separated) or $T' \sim \text{tens}$ (lattice), $\rho_{\text{lat}} = K_{\text{budget}}/K \geq 266.7/40.7 = 6.55 \geq 3.9$ and paired $\geq 6.55/2.872 = 2.28 > 1$. The floor obstruction is PROVED-removed; the SC-SCOPE all-orders endpoint LIFTS.

Grade and flip. The second-order floor obstruction (the SOLE named blocker) is now PROVED removed. The all-orders selection’s remaining input is the third-order inflation $1 + \max[R_s + R_q] = 2.872$, which is ESTIMATE-grade (R_s, R_q cited estimates) – the same basis the operator accepted for the $I \leq 10^{-3}$ feasibility, now extended uniformly to the endpoint. On the operator’s standing authorisation (lift if the reconciliation passes), SC-SCOPE is LIFTED: it is removed from B1’s active hypothesis set, $\{\text{H-LAYER}, \text{SC-SCOPE}\} \rightarrow \{\text{H-LAYER}\}$, with B1 tier UNCHANGED T6. The all-orders scope carries the explicit caveat that the third cumulant is controlled at estimate-grade (not rigorously proved).

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) “The constant map is unproved.” RESOLVED in v1.1 (§5): $K = \sum_{t \neq 0} w_t^2 / (\lambda' I)^2 \leq T'(M)$ is proved (Parseval + Lemma A $t \neq 0$ part) and verified ($K_{\text{floor}}/T' \leq 0.52$, $w_0 = I$). The remaining estimate-grade input is the third-order inflation (objection β), not the floor; the lift carries that caveat explicitly.

(β) “Does the third-order inflation also change with the tighter bound?” DISMISSED: the inflation 2.872 is the third-cumulant per-transfer cost (R_s, R_q), independent of the second-order additive-energy constant K ; only the floor ρ moves.

(γ) “ $T' \leq n_{\text{pack}}$ assumes the separated count; post-discharge the competitor may have more modes.”
 ADDRESSED: post-discharge the competitor is a lattice-shell subset with $T' \sim \text{tens}$ (R-026, measured ≤ 48), still < 67 ; and in the separated regime $T' \leq n_{\text{pack}} = 41$. Both close. The break-even $T' \leq 67$ has margin against both.

(δ) “Is $\rho = 2.58$ really reproduced?” Verified: $K_{\text{budget}}/K(n_{\text{pack}}) = 266.7/103.5 = 2.58$ matches the $\text{AddE}/\text{remargin} \times 2.6$ (scscope_floor_sharpening.py claim 1).

Quantitative sanity check (mandatory). $\rho_{\text{cons}} = 2.58$ (reproduces $\times 2.6$); $K_{\text{cons}} = 103.5 > 1 + n_{\text{pack}} = 41.7$ (bound loose); $\rho_{\text{lat}} \geq 6.4$ at $T' = n_{\text{pack}}$, paired ≥ 2.23 ; break-even $T' \leq 92/67$; lattice $T' \leq 48 \Rightarrow \text{paired} \geq 1.90$. Direction: tighter $K \Rightarrow \text{larger } \rho \Rightarrow \text{larger paired}$; all positive.

7. Result footer

Result ID:	R-029 / SC-SCOPE endpoint floor sharpening (candidate)
Precise statement:	The SC-SCOPE endpoint floor $\rho=2.58$ used the kappa-balanced bound $K(n_{\text{pack}})=8+4\sqrt{14}\sqrt{n_{\text{pack}}}=103.5$, which overshoots the Lemma-A additive-energy bound $1+T' \leq 1+n_{\text{pack}}=41.7$ at the small endpoint $n_{\text{pack}}=40.7$. Substituting $1+T'$ gives $\rho_{\text{lat}}=K_{\text{budget}}/(1+T') \geq 6.4$ (separated $T' \leq n_{\text{pack}}$), paired $=\rho_{\text{lat}}/2.872 \geq 2.23 > 1$ -- the all-orders endpoint CLOSES. Break-even $T' \leq 92$ (paired ≥ 1), $T' \leq 67$ ($\rho \geq 3.9$); lattice $T' \sim \text{tens}$ closes.
Scope:	SC-SCOPE all-orders endpoint ($I=2e-3$); the named 'sharper floor $\rho \geq 3.9$ ' route, now completed.
Dependencies:	robustness_mu2_step5b_remarg (floor); R-025 Lemma A (additive-energy bound $1+T'$); R-026 (lattice $T' \sim \text{tens}$); scscope-joint-pairing v1.0 (third-order inflation 2.872).
Evidence grade:	Reconciliation PROVED ($K \leq T'(M)$), verified 3/3). The floor obstruction is removed; SC-SCOPE all-orders endpoint LIFTS at estimate-grade third-order inflation. [superseded fields below kept for history] Residual: exact constant map kappa-balanced $K(n) \leftrightarrow \text{Lemma-A } 1+T'$ ($-4I^2 / \text{averaging}$). Verified numerically (5/5).
Reproduction command:	python codes/vacuum/scscope_floor_sharpening.py
Expected output:	5/5 PASS; $\rho_{\text{cons}} 2.58$; $K_{\text{cons}} 103.5 > 1+n_{\text{pack}} 41.7$; $\rho_{\text{lat}} \geq 6.4$ and paired ≥ 2.23 at $T'=n_{\text{pack}}$; break-even $T' \leq 92$.
Falsification gate:	The constant map giving an effective K_{lat} that keeps $\rho_{\text{lat}} < 3.9$ (T' effectively > 67 in the floor's normalisation) would leave the endpoint open; the reconciliation is the registered check.
Tier before / after:	ENACTED (operator-authorized on reconciliation): SC-SCOPE LIFTED, B1 hypotheses $\{\text{H-LAYER}, \text{SC-SCOPE}\} \rightarrow \{\text{H-LAYER}\}$, B1 T6 unchanged (all-orders scope, 3rd cumulant estimate-grade). B5 T5.
No-overclaim statement:	The floor obstruction is PROVED removed ($K \leq T'(M)$). The all-orders lift's third cumulant is ESTIMATE-grade (R_s, R_q), not rigorously proved -- so B1's all-orders scope carries that caveat. SC-SCOPE lifted (2nd-cumulant restriction removed); B1 tier UNCHANGED T6 on $\{\text{H-LAYER}\}$.
Next required action:	DONE (reconciliation passed; SC-SCOPE lifted). Mainline: B1 now rests on H-LAYER alone. Optional: promote the third-order inflation from estimate-grade to proved to make the all-orders scope rigorous.